

MUSHROOM CLASSIFICATION

TASK: TRAIN EDIBLE/POISONOUS CLASSIFIER →
MLFLOW LOGGING → DOCKERIZE → DEPLOY TO
RENDER WITH REST API.





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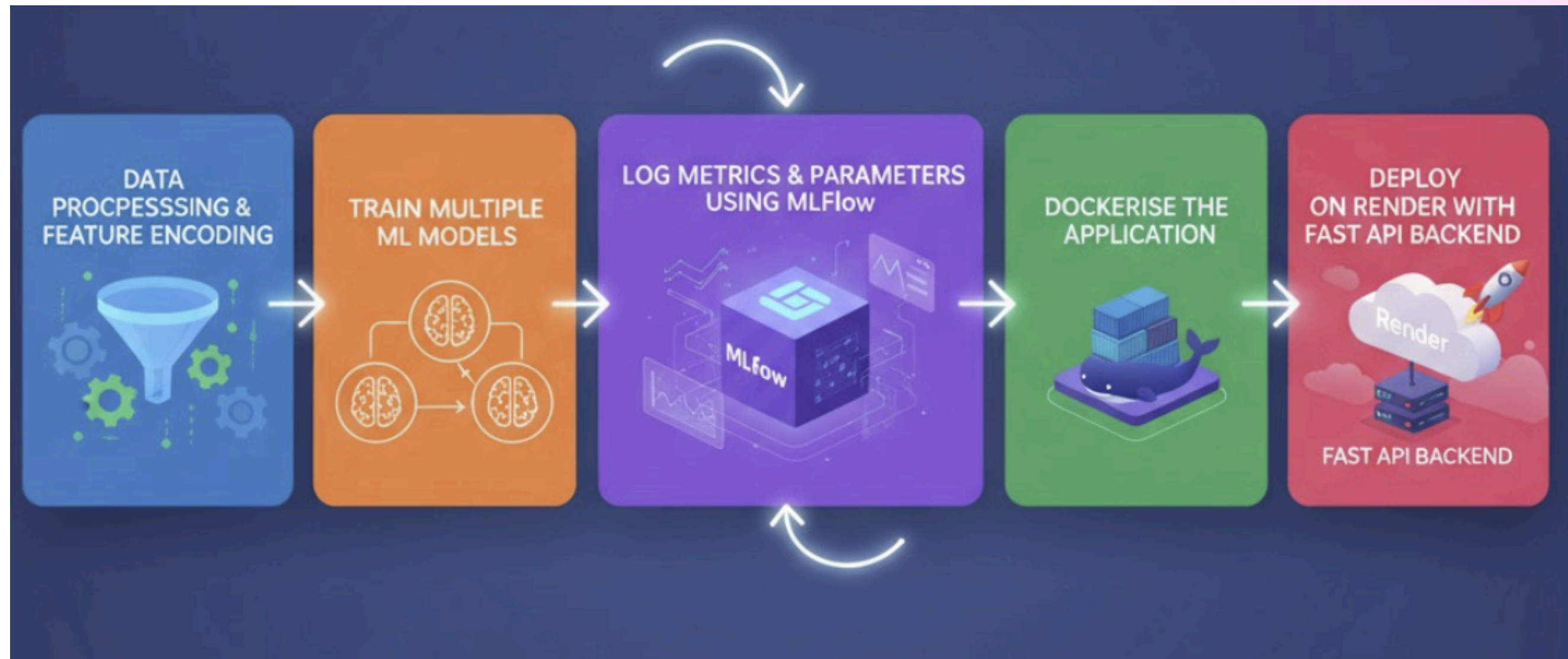
Srushti Moghe
24030242035



Vaisali Gupta
24030242074

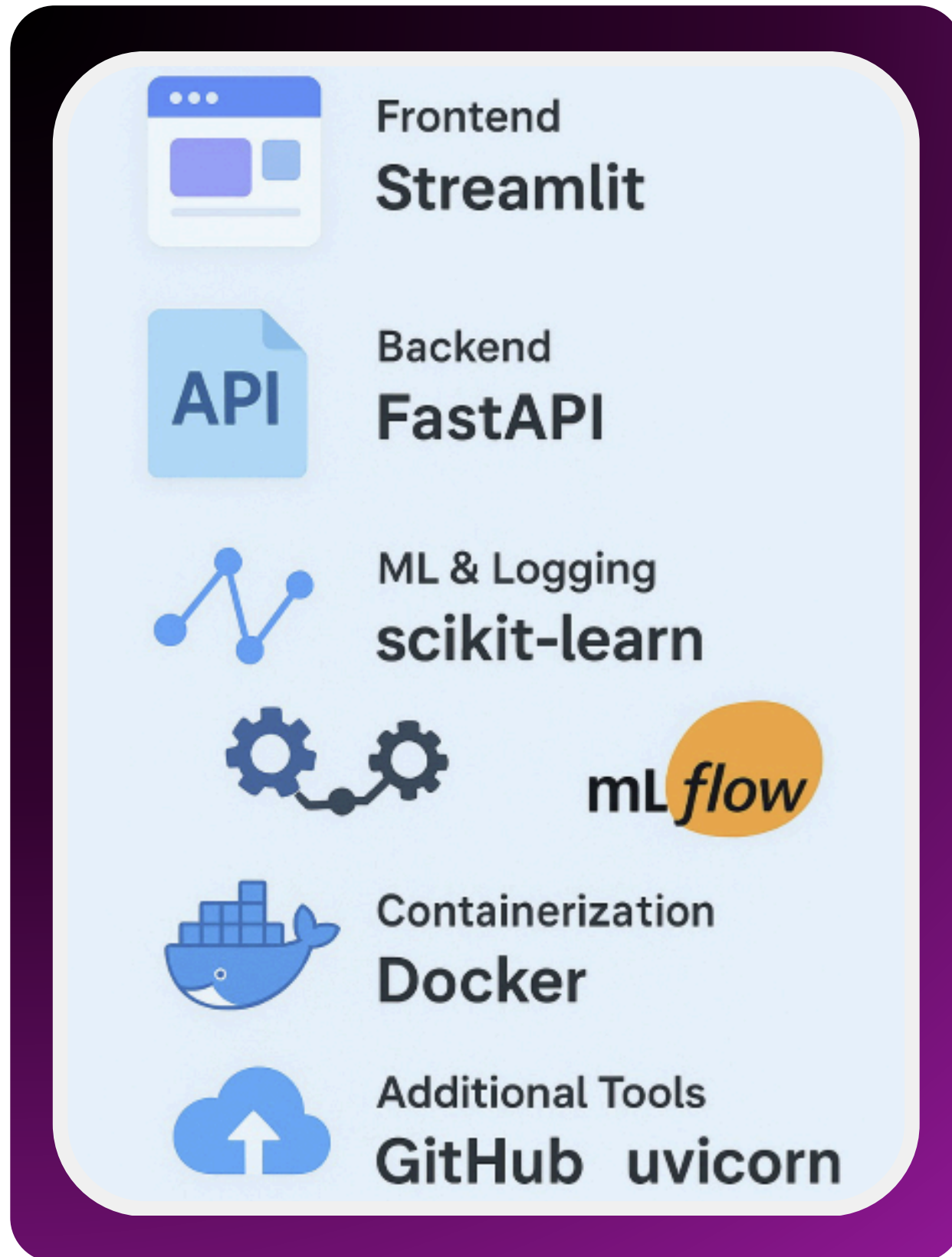
PROJECT OVERVIEW

Objective: Build a classifier to predict whether a mushroom is edible or poisonous.



Key Goal: Provide a user-friendly interface to predict mushrooms in real-time.

TECHNOLOGY STACK



- **Frontend:** Streamlit for interactive UI
- **Backend:** FastAPI for REST API endpoints
- **ML & Logging:** scikit-learn, ML Classification models, MLflow for model tracking
- **Containerization:** Docker
- **Deployment:** Render.com
- **Additional Tools:** GitHub for version control, uvicorn for API server

MODEL TRAINING & MLFLOW

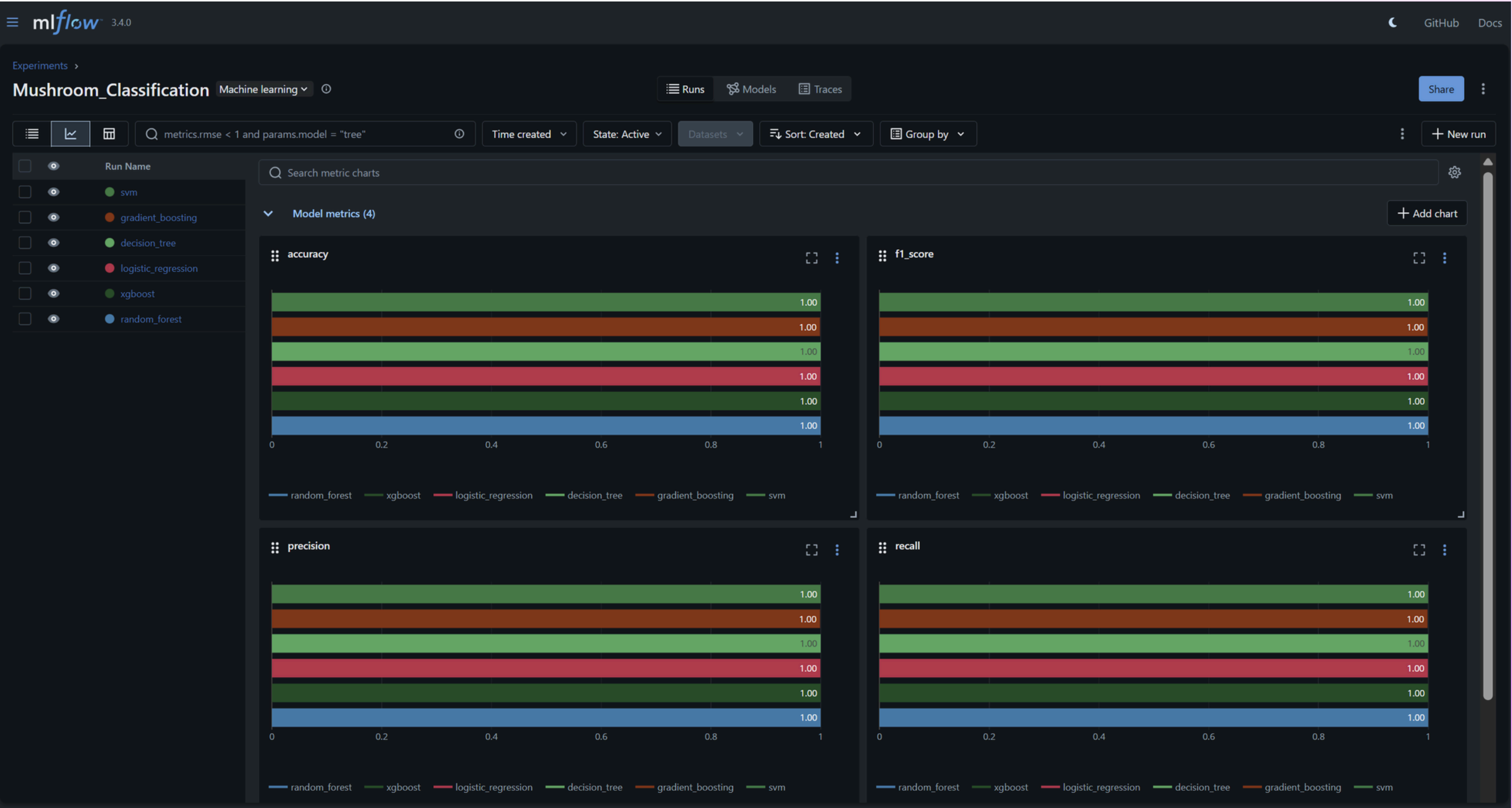
01 **Models Trained:**

- Random Forest
- XGBoost
- Support Vector Machine (SVM)
- Decision Tree
- Gradient Boosting
- Logistic Regression

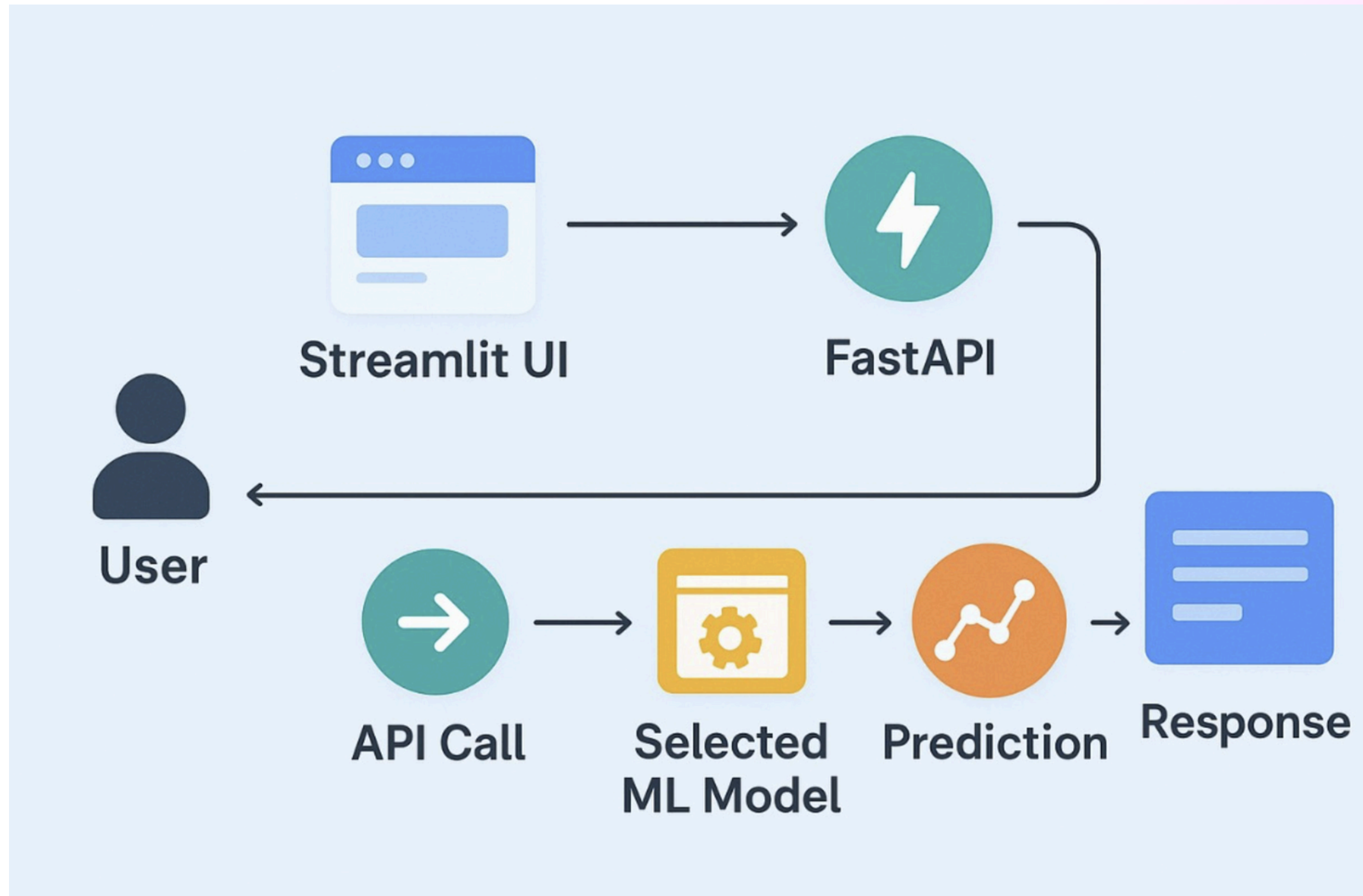
02 **MLflow Usage:**

- Logged metrics: Accuracy, Precision, Recall, F1-score
- Logged artifacts: Model files for deployment
- MLflow UI allowed easy comparison of model performance

MLFLOW UI



ARCHITECTURE & DEPLOYMENT



FAST API BACKEND



Mushroom Classification API

[/openapi.json](#)

1.0.0OAS 3.1

Predict whether a mushroom is edible ('e') or poisonous ('p').

default

GET / Root

POST /predict Predict Mushroom

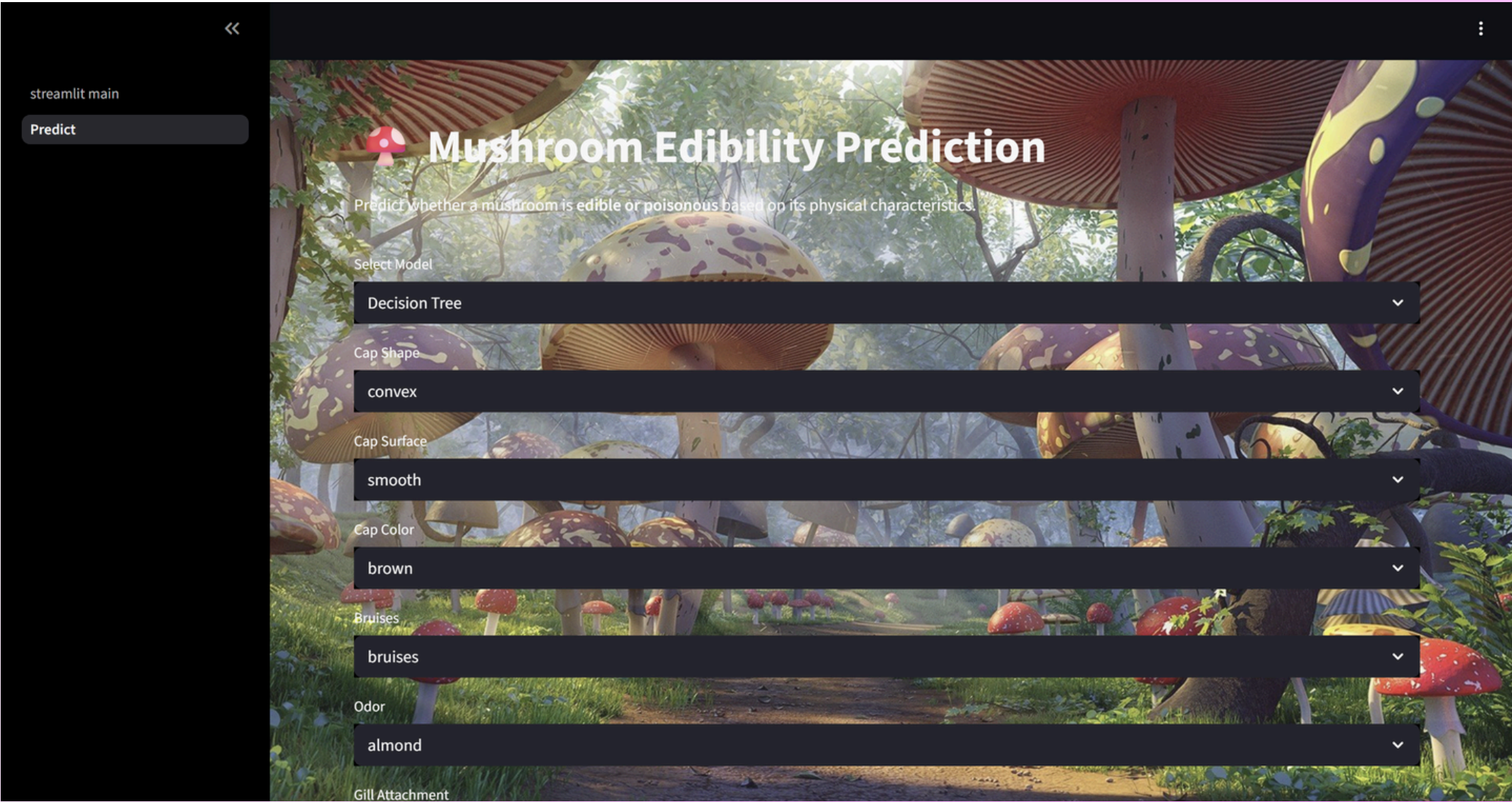
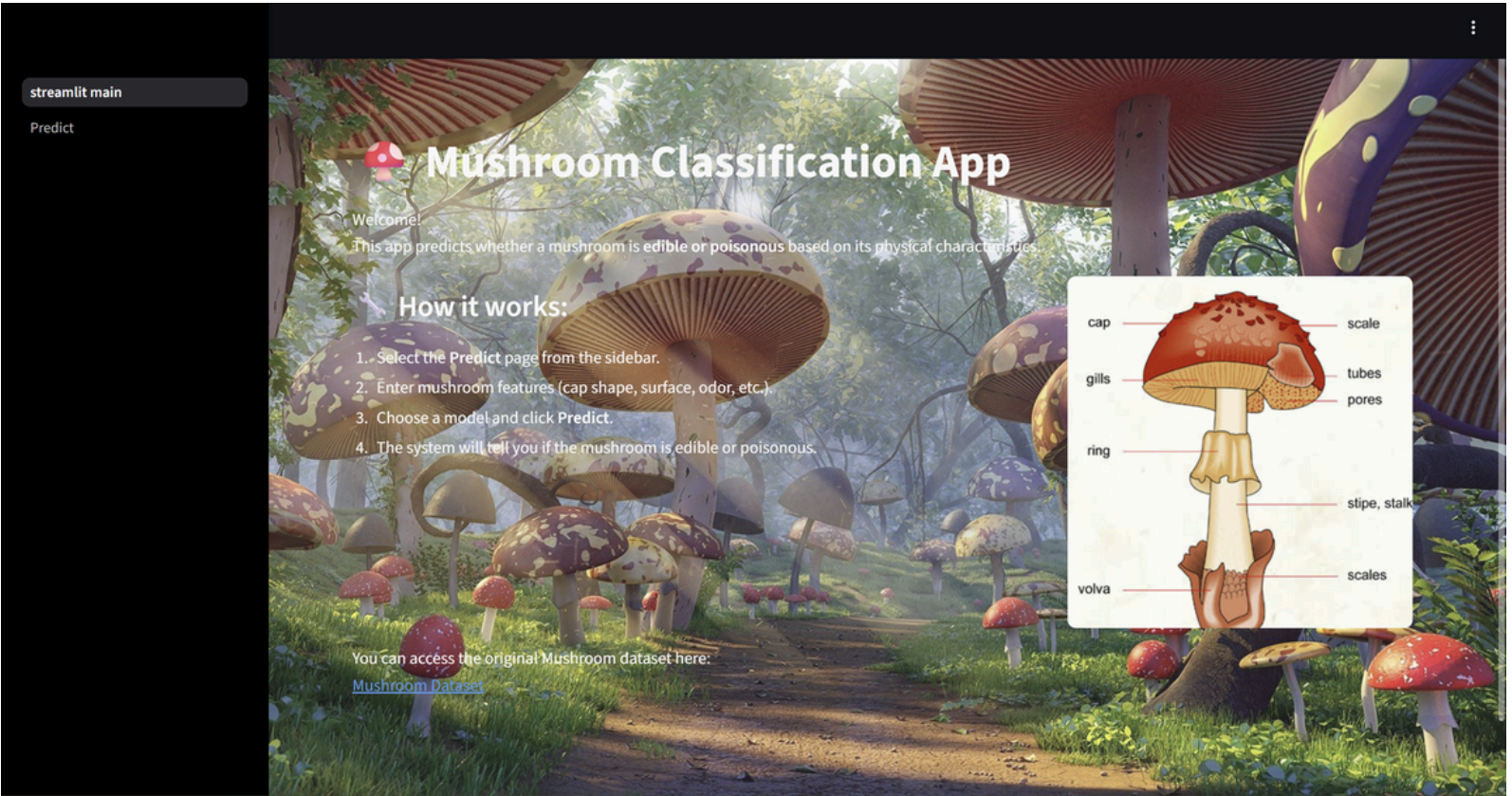
Schemas

HTTPValidationError > Expand all object

MushroomInput > Expand all object

ValidationError > Expand all object

STREAMLIT FRONTEND



DOCKER DESKTOP

docker.desktop

PERSONAL

Search

Ctrl+K

3

H

Ask Gordon

Containers

Images

Volumes

Kubernetes

Builds

Models

MCP Toolkit

Docker Hub

Docker Scout

Extensions

Containers

Container CPU usage

0.34% / 1600% (16 CPUs available)

Container memory usage

182.4MB / 7.36GB

Show charts

Search

Only show running containers

	Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
☐	objective_davinci	b06a051740dc	mushroom-app	8000:8000	0%	6 hours ago	
☐	reverent_hypatia	b0e8abdf4992	mushroom-app	8000:8000	0%	6 hours ago	
☐	quirky_williams	e04dad40f1a8	mushroom-app	8000:8000	0%	6 hours ago	
☐	festive_roentgen	ea2dac44df33	mushroom-app	8000:8000	0%	6 hours ago	
☐	festive_hopper	dc22810108ee	mushroom-app	8000:8000	0%	6 hours ago	
☐	sad_feynman	12bf5c263ff7	mushroom-app	8000:8000	0%	6 hours ago	
☐	dazzling_taussig	5e617736a28e	mushroom-app	8000:8000	0%	6 hours ago	
☐	xenodochial_shirle	6059b2d5862d	mushroom-app	8000:8000	0%	6 hours ago	
☐	hopeful_hertz	0533cec195cc	mushroom-app	8000:8000	0%	6 hours ago	
☐	nifty_swirles	c4617e25e4e2	mushroom-app	8000:8000	0%	6 hours ago	
☐	epic_agnesi	0bbe84c62f08	mushroom-app	8000:8000	0%	6 hours ago	
☐	compassionate_g	57e46687a9f7	myapp:latest	8000:8000	0.3%	24 minutes ago	

Showing 26 items

Engine running

RAM 2.06 GB CPU 9.82% Disk: 31.26 GB used (limit 1006.85 GB)

Terminal

New version available

GITHUB REPOSITORY

hariharan-codes / Mushroom_Classification_Project

🔍 Type / to search

<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

Mushroom_Classification_Project

Public

📌 Pin

👁 Watch 0

🍴 Fork 0

★ Star 0

🔗 main

🔗 1 Branch

🏷 0 Tags

🔍 Go to file

Add file

<> Code

hariharan-codes

Update Dockerfile and start.sh for Render deployment

196c944 · 39 minutes ago

🕒 4 Commits

📁 data	Initial commit - Dockerized Mushroom Classification Project	3 hours ago
📁 frontend	Reorganize Streamlit multi-page app for Render	1 hour ago
📁 mlruns	Initial commit - Dockerized Mushroom Classification Project	3 hours ago
📁 models	Initial commit - Dockerized Mushroom Classification Project	3 hours ago
📁 myenv	Initial commit - Dockerized Mushroom Classification Project	3 hours ago
📁 src	Initial commit - Dockerized Mushroom Classification Project	3 hours ago
📄 Dockerfile	Update Dockerfile and start.sh for Render deployment	48 minutes ago
📄 README.md	Initial commit - Dockerized Mushroom Classification Project	3 hours ago
📄 requirements.txt	Initial commit - Dockerized Mushroom Classification Project	3 hours ago
📄 start.sh	Update Dockerfile and start.sh for Render deployment	39 minutes ago
📄 train.py	Initial commit - Dockerized Mushroom Classification Project	3 hours ago

About

No description, website, or topics provided.

📶 Activity

★ 0 stars

👁 0 watching

🍴 0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Languages

Python 92.6%

Cython 2.7%

C 0.6%

C++ 3.3%

JavaScript 0.6%

CSS 0.1%

Other 0.1%

RENDER DEPLOYMENT

The screenshot displays the Render dashboard interface. The top navigation bar includes 'My Workspace', the project name 'Mushroom_Classification_Project', a search bar, and buttons for '+ New', 'Upgrade', and user profile. The left sidebar contains navigation links: Dashboard, Mushrooms, Events (highlighted), Settings, MONITOR (Logs, Metrics), and MANAGE (Environment, Shell, Scaling, Previews, Disks, Jobs, Changelog, Invite a friend, Contact support, Render Status). The main content area shows a deployment event for 'September 27, 2025 at 6:07 PM' with a 'Live' status. Below this, a log viewer displays the deployment process, including a warning about the free instance spinning down and a message about no open ports detected. The logs also show the service is live and provide the primary URL: <https://mushroom-classification-project-nyzk.onrender.com>. A warning message from XGBoost is also visible, advising on model saving and loading. At the bottom, a footer suggests using the Render CLI, MCP Server, or log stream integration for better log management.

My Workspace

Mushroom_Classification_Project

Search

+ New Upgrade

Dashboard

Mushroom_Classification_Pr...

Events

Settings

MONITOR

Logs

Metrics

MANAGE

Environment

Shell

Scaling

Previews

Disks

Jobs

Changelog

Invite a friend

Contact support

Render Status

September 27, 2025 at 6:07 PM Live

196c944 Update Dockerfile and start.sh for Render deployment

All logs Search

Live tail GMT+5:30

Sep 27 06:19:16 PM

Sep 27 06:19:17 PM ==> No open ports detected, continuing to scan...

Sep 27 06:19:17 PM ==> Docs on specifying a port: <https://render.com/docs/web-services#port-binding>

Sep 27 06:19:24 PM ==> Your service is live 🎉

Sep 27 06:19:25 PM ==>

Sep 27 06:19:25 PM ==> //////////////////////////////////////

Sep 27 06:19:25 PM ==>

Sep 27 06:19:25 PM ==> Available at your primary URL <https://mushroom-classification-project-nyzk.onrender.com>

Sep 27 06:19:25 PM /usr/local/lib/python3.9/site-packages/xgboost/core.py:158: UserWarning: [12:49:25] WARNING: /workspace/src/collective/./data/./common/error_msg.h:80: If you are loading a serialized model (like pickle in Python, RDS in R) or

Sep 27 06:19:25 PM configuration generated by an older version of XGBoost, please export the model by calling

Sep 27 06:19:25 PM 'Booster.save_model' from that version first, then load it back in current version. See:

Sep 27 06:19:25 PM

Sep 27 06:19:25 PM https://xgboost.readthedocs.io/en/stable/tutorials/saving_model.html

Sep 27 06:19:25 PM

Sep 27 06:19:25 PM for more details about differences between saving model and serializing.

Sep 27 06:19:25 PM

Need better ways to work with logs? Try the [Render CLI](#), [Render MCP Server](#), or set up a log stream integration [↗](#)

Link: <https://mushroom-classification-project-nyzk.onrender.com>

THANK YOU